

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.impute import SimpleImputer
from sklearn.pipeline import Pipeline
import os
```

✓ 0.0s

```
def load_data(path):
    print("Loading data...")
    return pd.read_csv(path)
```

✓ 0.0s

```
def preprocess_data(df):
    print(" Preprocessing data...")

    imputer = SimpleImputer(strategy='mean')
    df[['Age']] = imputer.fit_transform(df[['Age']])
    df[['Income']] = imputer.fit_transform(df[['Income']])

    # Encode categorical features
    le = LabelEncoder()
    df[['Gender']] = le.fit_transform(df[['Gender']])
    df[['Purchase']] = le.fit_transform(df[['Purchase']])

    # Feature scaling
    scaler = StandardScaler()
    df[['Age', 'Income']] = scaler.fit_transform(df[['Age', 'Income']])

    return df
```

✓ 0.0s

```
def split_and_save(df, output_folder="processed_data"):
    print("Splitting and saving data...")
    X = df.drop('Purchase', axis=1)
    y = df['Purchase']

    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

    os.makedirs(output_folder, exist_ok=True)
    X_train.to_csv(f"{output_folder}/X_train.csv", index=False)
    X_test.to_csv(f"{output_folder}/X_test.csv", index=False)
    y_train.to_csv(f"{output_folder}/y_train.csv", index=False)
    y_test.to_csv(f"{output_folder}/y_test.csv", index=False)

    print(" Data saved in 'processed_data' folder.")
    return X_train, X_test, y_train, y_test
```

✓ 0.0s

```
def run_etl_pipeline(file_path):
    data = load_data(file_path)
    processed = preprocess_data(data)
    split_and_save(processed)
```

✓ 0.0s

```

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.impute import SimpleImputer
from sklearn.datasets import fetch_california_housing
import os

def load_or_download_data(csv_path="california_housing_train.csv"):
    if os.path.exists(csv_path):
        print(f"Found dataset at: {csv_path}")
        return pd.read_csv(csv_path)
    else:
        print("Downloading California housing dataset...")
        housing = fetch_california_housing(as_frame=True)
        df = housing.frame
        df.to_csv(csv_path, index=False)
        print(f"Saved dataset to: {csv_path}")
        return df

def preprocess_data(df):
    print("Preprocessing data...")
    num_cols = df.select_dtypes(include=['int64', 'float64']).columns
    cat_cols = df.select_dtypes(include=['object']).columns

    # Handle missing numeric values
    if len(num_cols) > 0:
        imputer_num = SimpleImputer(strategy='mean')
        df[num_cols] = imputer_num.fit_transform(df[num_cols])

    # Handle missing categorical values + encoding
    if len(cat_cols) > 0:
        imputer_cat = SimpleImputer(strategy='most_frequent')
        df[cat_cols] = imputer_cat.fit_transform(df[cat_cols])
        for col in cat_cols:
            le = LabelEncoder()
            df[col] = le.fit_transform(df[col])

    # Scale numeric features
    if len(num_cols) > 0:
        scaler = StandardScaler()
        df[num_cols] = scaler.fit_transform(df[num_cols])

    print("Preprocessing completed.")
    return df

def split_and_save(df, output_folder="processed_data"):
    print("Splitting and saving data...")
    X = df.iloc[:, :-1]
    y = df.iloc[:, -1]

    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
    os.makedirs(output_folder, exist_ok=True)
    X_train.to_csv(f"{output_folder}/X_train.csv", index=False)
    X_test.to_csv(f"{output_folder}/X_test.csv", index=False)
    y_train.to_csv(f"{output_folder}/y_train.csv", index=False)
    y_test.to_csv(f"{output_folder}/y_test.csv", index=False)

    print("Data saved in 'processed_data' folder.")
    return X_train, X_test, y_train, y_test

def run_etl_pipeline():
    data = load_or_download_data()
    processed = preprocess_data(data)
    split_and_save(processed)

if __name__ == "__main__":
    run_etl_pipeline()

```

- 📶 Downloading California housing dataset...
- ✅ Saved dataset to: california_housing_train.csv
- 🔪 Preprocessing data...
- ✅ Preprocessing completed.
- 📁 Splitting and saving data...
- ✅ Data saved in 'processed_data' folder.